Franklin Diaz

Lecturer — Information Systems | Principal Security Consultant

University of Colorado Denver Business School • Palo Alto Networks • franklin.diaz@ucdenver.edu • bitsmasher.net • linkedin.com/in/fdiaz • github.com/devsecfranklin

Education

Master of Science in Computer Security & **Master of Science in Information Systems**
DePaul University, Chicago, IL (2007) & Northwestern University, Evanston, IL (2004)

Bachelor of Science in Computer Science
Roosevelt University, Chicago, IL (2002)

Academic Appointments

Lecturer, Information Systems Discipline — **Sep 2025 – Present**
University of Colorado Denver Business School — Denver, CO

- Teaches **ISMG 2050: Business Problem Solving Tools**, introducing IT infrastructure, data management systems, analytical modeling, and computational business problem solving.

Adjunct Faculty — Information Systems — **2006 – 2011**
McHenry County College — Crystal Lake, IL

- Taught undergraduate courses in Linux OS fundamentals, shell programming, enterprise productivity suites, and Microsoft Windows operating systems.

Professional Experience

Principal Security Consultant / Extended Expertise — **Jan 2020 – Present**
Palo Alto Networks — Remote

- Lead architecture and deployment of Palo Alto Networks security solutions for enterprise clients executing multi-cloud migrations (GCP, OCI, AWS, Azure).
- Architect scalable infrastructure using Infrastructure as Code (IaC), aligning mission-critical environments to rigorous zero-trust reference architectures.
- Serve as technical advisor identifying policy and architectural gaps, delivering remediation roadmaps, and advising executive stakeholders post-incident.

Security Engineer (Detection & Response Engineering) — **Oct 2014 – Jan 2020**
Salesforce — Bellevue, WA

- Built high-throughput security telemetry collection systems processing **over 1 billion security events per day** across global enterprise infrastructure.
- Engineered and maintained analytical data pipelines in a FedRAMP High GovCloud environment on AWS.
- Developed distributed packet capture infrastructure, CI/CD automation, and integrated regression test suites across multi-repository toolsets.

Principal Research Software Engineer — **Aug 2012 – Jan 2014**
Role: Security Engineer & Test Architect | *Indiana University (SDN Lab) — Bloomington, IN*

- Directed testing strategies and test case engineering for Software Defined Networking (OpenFlow) infrastructure.
- Developed web-based telemetry and management tools for the GENI/GEMINI national experimental network testbed.
- Assisted undergraduate computer networking courses: led discussions, developed coursework, and proctored examinations.

Research & Development Engineer — **Apr 2011 – Jul 2012**
Location: Espoo, Finland & Arlington Heights, IL | *Nokia Siemens Networks*

- Conducted cellular base station product development, firmware integration, and multi-subsystem verification for CDMA telecom systems.

Senior Software Engineer — **Jan 1998 – Mar 2011**
Cellular Networks & Space Sector — 13-Year Core Engineering Tenure | *Motorola Solutions*

- Engineered CDMA cellular base station infrastructure and embedded subsystems using C and C++.
- Partnered directly with premier carrier account teams including Sprint, ALLTEL, Lucent, Pelephone, and NTT on systems integration and network testing.

- Led cross-functional system integration efforts and acceptance testing across distributed product feature teams.

Projects & Open Research

- **Cloud Native Security Lab** (Zenodo Record 5918861): Complete open-source cloud-native security research lab deployed in Google Cloud Platform.
- **project emerald**: Minecraft schematic generation utilizing local AI Edge TPU hardware acceleration.
- **Peirates**: Contributor to open-source Kubernetes penetration testing and attack-surface evaluation tool.

Publications, Talks & Community Service

- **Conference Security Track Volunteer:** BSidesLV.
- **Technical Author & Researcher:** Decentralized protocols, autonomous agent runtime security, and DevSecOps at bitsmasher.net/research.

Technical Skills & Competencies

Languages: & Python, SQL, Go, JavaScript, Bash, C/C++, Java, PERL
Frameworks & Libraries: & Pydantic, FastAPI, Django, Leaflet, LINQ
Cloud & Platforms: & AWS, GCP, Azure, OCI
Tools & Infrastructure: & Git, JIRA, Kafka, Redis, Docker, Kubernetes, Terraform, Airflow, Jenkins, LaTeX
Concepts & Practices: & DevSecOps, Infrastructure as Code, REST/XML APIs, Object-Oriented Design, TDD, Multi-Cloud Architecture